

Week 6 - Windows Forensics

I. Multiple Choice Questions (MCQ)

1. Which tool is specifically designed for live data acquisition of registry data from a running Windows system?
 - a) RegRipper
 - b) KAPE
 - c) Registry Explorer
 - d) Regedit
2. Which tool allows investigators to extract files from either a disk image or a live system by mounting the image or drive?
 - a) FTK Imager
 - b) Word
 - c) Wireshark
 - d) Nmap
3. Which Windows path stores the main offline registry hives such as SYSTEM, SOFTWARE, SAM, SECURITY, and DEFAULT?
 - a) C:\Users\Public
 - b) C:\Windows\AppCompat\Programs
 - c) C:\Windows\System32\Config
 - d) C:\Program Files\Registry
4. Which registry analysis tool can load multiple hives at the same time and merge transaction log data into a cleaner, more up-to-date view?
 - a) Registry Explorer
 - b) Registry Viewer
 - c) Event Viewer
 - d) FTK Imager
5. What is the main limitation of the basic Registry Viewer compared with Registry Explorer?
 - a) It cannot open registry hives
 - b) It only loads one hive at a time and does not account for transaction logs
 - c) It only works on Linux
 - d) It cannot display key values
6. What is the primary function of RegRipper?
 - a) Acquire RAM from live systems

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- b) Modify registry entries during analysis
 - c) Extract forensic data from registry hives into a text report
 - d) Encrypt registry hives
7. Which registry key is used to identify the Windows OS version?
- a) SOFTWARE\Microsoft\Windows NT\CurrentVersion
 - b) SYSTEM\CurrentControlSet\Control\ComputerName
 - c) NTUSER.DAT\Software\Microsoft\Windows\Shell\Bags
 - d) SYSTEM\CurrentControlSet\Enum\USBSTOR
8. Which location identifies the active control set on a live system?
- a) SOFTWARE\Microsoft\Windows\CurrentVersion\Run
 - b) SYSTEM\Select\Current
 - c) NTUSER.DAT\Software\Microsoft\Office
 - d) SYSTEM\CurrentControlSet\Enum\USB
9. Which registry path contains the computer name?
- a) SYSTEM\CurrentControlSet\Control\ComputerName\ComputerName
 - b) SOFTWARE\Microsoft\Windows NT\CurrentVersion
 - c) SYSTEM\CurrentControlSet\Services\bam\UserSettings
 - d) NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\TypedPaths
10. Which registry path stores timezone information useful for timeline analysis?
- a) SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList
 - b) SYSTEM\CurrentControlSet\Control\TimeZoneInformation
 - c) NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs
 - d) SYSTEM\CurrentControlSet\Enum\USBSTOR
11. Which registry path contains network interface information?
- a) SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces
 - b) SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce
 - c) NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\WordWheelQuery
 - d) SAM\Domains\Account\Users
12. Which registry key stores information about past networks?
- a) SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList\Signatures
 - b) SYSTEM\CurrentControlSet\Services\dsm\UserSettings
 - c) NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist
 - d) SOFTWARE\Microsoft\Windows Portable Devices\Devices
13. Which registry hive stores local user account information such as identifiers, account creation data, usernames, groups, and descriptive comments?
- a) SECURITY
 - b) DEFAULT

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- c) SAM
 - d) USRCLASS.DAT
14. Which registry location is one of the standard autorun paths for startup programs?
- a) SOFTWARE\Microsoft\Windows\CurrentVersion\Run
 - b) SYSTEM\CurrentControlSet\Control\TimeZoneInformation
 - c) NTUSER.DAT\Software\Microsoft\Windows\Shell\Bags
 - d) SYSTEM\CurrentControlSet\Enum\USB
15. What does a service Start value of 0x02 typically indicate?
- a) The service is disabled
 - b) The service starts at boot
 - c) The service runs only manually
 - d) The service was deleted
16. Which artifact is commonly used to identify recently opened files from Windows Explorer?
- a) UserAssist
 - b) RecentDocs
 - c) ShimCache
 - d) BAM
17. Which key can help identify recently opened PDF files specifically?
- a) NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\pdf
 - b) SYSTEM\CurrentControlSet\Enum\USBSTOR
 - c) SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList
 - d) SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces
18. Which registry area stores recent Microsoft Office file history?
- a) NTUSER.DAT\Software\Microsoft\Office\VERSION\UserMRU\LiveID_####\FileMRU
 - b) SYSTEM\CurrentControlSet\Control\ComputerName
 - c) SOFTWARE\Microsoft\Windows Portable Devices\Devices
 - d) SAM\Domains\Account\Users
19. Which artifact stores folder view preferences and can help show knowledge of accessed folders?
- a) TypedPaths
 - b) ShellBags
 - c) UserAssist
 - d) USBSTOR
20. Which registry keys track locations used in file open/save dialogs?
- a) OpenSavePIDMRU and LastVisitedPidlMRU
 - b) UserAssist and ShimCache

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- c) Run and RunOnce
 - d) SAM and SECURITY
21. Which key records paths typed in the Windows Explorer address bar?
- a) TypedPaths
 - b) WordWheelQuery
 - c) BAM
 - d) AppCompatCache
22. Which key stores Windows Explorer search terms?
- a) RecentDocs
 - b) WordWheelQuery
 - c) ShellBags
 - d) USB
23. Which artifact tracks programs launched through Windows Explorer for statistical purposes?
- a) UserAssist
 - b) USBSTOR
 - c) NetworkList
 - d) TypedPaths
24. Which statement best describes ShimCache/AppCompatCache?
- a) It records USB volume names
 - b) It tracks application compatibility and stores executable name, size, and last modified time information
 - c) It stores email attachments
 - d) It only tracks browser history
25. What additional evidence can AmCache provide beyond basic execution traces?
- a) Execution path, installation/execution/deletion times, and SHA1 hashes
 - b) Only printer settings
 - c) Only network passwords
 - d) Only screen resolution values
26. What do BAM and DAM mainly help track?
- a) Background application activity and power-related activity information
 - b) USB serial numbers only
 - c) Deleted registry keys only
 - d) Browser cookies only
27. Which registry path is commonly used to identify connected USB storage devices?
- a) SYSTEM\CurrentControlSet\Enum\USBSTOR
 - b) NTUSER.DAT\Software\Microsoft\Office\VERSION

- c) SOFTWARE\Microsoft\Windows\CurrentVersion\Run
 - d) SAM\Domains\Account\Users
28. Which registry path stores general USB device vendor ID, product ID, and version details?
- a) SYSTEM\CurrentControlSet\Enum\USB
 - b) SOFTWARE\Microsoft\Windows NT\CurrentVersion
 - c) NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs
 - d) SYSTEM\CurrentControlSet\Services\bam\UserSettings
29. Where can the volume name of a connected USB device be found?
- a) SOFTWARE\Microsoft\Windows Portable Devices\Devices
 - b) SYSTEM\Select\Current
 - c) NTUSER.DAT\Software\Microsoft\Windows\Shell\BagMRU
 - d) SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run
30. Which detail is commonly included in USB hunting on Windows?
- a) Volume GUID and last mount point or drive letter
 - b) Only desktop wallpaper history
 - c) Only browser autofill entries
 - d) Only printer spool paths

II. True/False Questions

1. KAPE can be used to acquire registry data from a live Windows system.
2. Registry Explorer can load multiple hives and incorporate transaction logs.
3. RegRipper produces a text report from forensically important registry keys and values.
4. The main offline registry hives are stored in C:\Users\<username>.
5. CurrentControlSet is a volatile control set used while the machine is live.
6. ControlSet001 always represents the last known good configuration.
7. Knowledge of the local timezone is useful when building a forensic timeline.
8. The SAM hive can contain user account information.
9. The Run and RunOnce keys may contain startup entries.
10. A service with Start value 0x02 typically starts at boot.
11. RecentDocs helps identify recently opened files.

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12. ShellBags can reveal folder view preferences and user interaction with folders.
 13. TypedPaths stores Windows Explorer search queries.
 14. WordWheelQuery stores Windows Explorer search history.
 15. UserAssist records all programs executed from both Windows Explorer and the command line.
 16. ShimCache is also called AppCompatCache.
 17. AmCache may store SHA1 hashes of executed programs.
 18. BAM and DAM are related to background activity and Modern Standby behavior.
 19. USBSTOR can help identify connected USB storage devices.
 20. SOFTWARE\Microsoft\Windows Portable Devices\Devices can reveal a connected drive's volume name.

III. Fill in the Blanks

1. The process of copying or imaging data for forensic analysis is called _____.
2. The live acquisition tool used to collect registry data from a running system is _____.
3. The main offline registry hives are commonly stored in _____.
4. The tool that can load multiple hives and merge transaction logs is _____.
5. The utility that takes a registry hive as input and outputs a sequential text report is _____.
6. The registry key used to determine the operating system version is _____.
7. The value that identifies the active control set is found under _____.
8. The registry path used to find a computer's timezone settings is _____.
9. The registry path used to find network interface information is _____.
10. The hive that stores local account information is _____.
11. The key commonly used to track recently opened files is _____.
12. The Office-related MRU area discussed in the lecture is _____.
13. Folder layout and folder-view evidence can be found in _____.
14. The open/save dialog MRU locations include _____ and _____.
15. Paths typed into the Windows Explorer address bar are stored in _____.
16. Search terms typed in Windows Explorer are stored in _____.
17. The artifact that tracks programs launched through Windows Explorer is _____.
18. The compatibility-related execution artifact in the SYSTEM hive is _____.
19. The artifact located at C:\Windows\AppCompat\Programs\Amcache.hve is _____.
20. Two important registry locations for USB identification are _____ and _____.

IV. Short Answer Questions

1. Why is it recommended to image a live system or copy the required data before deep forensic analysis?
2. How does Registry Explorer improve on the basic Registry Viewer?
3. What is the forensic value of RegRipper output?
4. What kinds of system details can be collected from CurrentControlSet-related paths?
5. Why is timezone information important in forensic investigations?
6. What kinds of account-related information can be recovered from the SAM hive?
7. How can RecentDocs help an investigator?
8. What is the forensic value of ShellBags?
9. What is the difference between TypedPaths and WordWheelQuery?
10. Why might UserAssist fail to show some program executions?
11. Compare the forensic value of ShimCache and AmCache.
12. What do BAM and DAM contribute to program execution analysis?
13. How can investigators use USBSTOR and USB together during USB device analysis?
14. What types of USB details should an investigator try to recover?
15. How can file and folder artifacts be correlated to strengthen an investigation?

V. Essay Questions

1. Explain how the lecture's acquisition methods and tools support sound Windows forensic practice.
2. Discuss how registry analysis supports collection of system information and account information during an investigation.
3. Explain how files-and-folders artifacts can reveal user knowledge, access, and activity on a Windows system.
4. Compare UserAssist, ShimCache, AmCache, and BAM/DAM as sources of execution evidence.
5. Explain how USB device artifacts can be used to reconstruct external device usage on a Windows system.

VI. Scenario-based Questions

1. A workstation is still powered on during an internal investigation. Describe a sound first-response approach for acquiring relevant Windows forensic data, including which tools and artifact sources should be prioritized.
2. An analyst needs to determine the computer name, timezone, active control set, past networks, and startup items from a suspect machine. Explain which registry areas should be examined and what each contributes.
3. A user denies opening any confidential PDF files. Describe how an investigator could use Windows file-and-folder artifacts to test that claim.
4. A company suspects unauthorized software execution on a workstation. Explain how an investigator could combine UserAssist, ShimCache, AmCache, and BAM/DAM to assess what ran and how.
5. An employee is suspected of using a USB device to copy company data. Describe a structured registry-based approach to identify the device, its volume name, connection history, and user association.
6. A forensic examiner wants to build a stronger timeline of suspicious activity by combining system, file, execution, and USB artifacts. Propose a workflow that covers all four areas.